Requirement

| | |
|---|---|
| **Date** | 30 September 2026 |
| **Team ID** | 6ab4dd7a6f290be8a6b49f92 |
| **Project Name** | **Script-Controlled ACL – Restrict Record Access Based on Field Value** |
| **Maximum Marks** | 3 Marks |

## Requirement

The project **"Script-Controlled ACL – Restrict Record Access Based on Field Value"** requires a ServiceNow Developer Instance to implement field-based record-level access control. The main requirement is to ensure that users can access Institution Details records according to their assigned roles and the value of the Branch field.

### 1. Platform Requirement

The project shall be developed and tested using a **ServiceNow Developer Instance**. The ServiceNow **Application Navigator** shall be used to access the required modules, including Users, Roles, Tables, and Access Controls.

### 2. User and Role Requirements

A dedicated user shall be created in the ServiceNow instance for testing the access-control functionality. The project requires the creation of four roles:

- **BB1**
- **BB2**
- **BB3**
- **BB4**

The required roles shall be assigned to the project user. These roles are used to control different operations on the Institution Details table.

### 3. Institution Details Table Requirements

A custom table named **Institution Details** shall be created. The table shall contain the required fields for storing student and institutional information. The major fields include:

- Student ID
- Student Name
- Faculty Name
- Branch

- Email
- Phone Number
- Description

The Student Name and Faculty Name fields shall use appropriate reference data types where required. The Branch field shall be configured as a choice field.

## 4. Branch Choice Requirements

The **Branch** field shall contain the following choices:

- **ECE – Electronics and Communication Engineering**
- **EEE – Electrical and Electronics Engineering**
- **CSC – Computer Science**

These branch values are required for implementing field-based access restrictions.

## 5. Sample Data Requirements

The Institution Details table shall contain sample records representing students from different branches. The project demonstration includes records such as:

- Student 100 – ECE
- Student 200 – EEE
- Student 300 – CSC

These records are required to verify whether the ACL restrictions operate correctly.

## 6. Access Control Requirements

The project shall implement **Access Control Lists (ACLs)** for the following operations:

1. **Read ACL** – Controls which records the user can view.
2. **Create ACL** – Controls the ability to create new records.
3. **Write ACL** – Controls the ability to modify records.
4. **Delete ACL** – Controls the ability to delete records.

The Read ACL shall include a role requirement and a branch-based condition. The project shall use a **script-controlled authorization mechanism** to provide additional access control based on the value of the Branch field.

## 7. Security and Validation Requirements

The system shall restrict unauthorized access to records belonging to other branches. For example, the configured user with the required role and authorization should be able to view the permitted **EEE** records, while records belonging to ECE and CSC should not be displayed to that user.

Requirement

The system shall also allow appropriate administrative users to access the required records for management and verification purposes.

## 8. Expected Outcome

The completed system shall demonstrate that ServiceNow ACLs can be combined with roles, conditions, and scripts to implement **field-based record-level security**. The final implementation must successfully distinguish between authorized and unauthorized records and control Read, Create, Write, and Delete operations according to the configured security rules.